ibm_brisbane

Table 1: Similarity matrix for ibm_brisbane, five-qubit circuit with depth 10, and optimization level 0

	***ibm_brisbane* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeBrisbane***	**Noise Model From CSV**
***ibm_brisbane* QPU**	100%	65.619%	65.915%	64.392%	67.101%
Simulator Instance From QPU	65.619%	100%	96.159%	96.509%	94.531%
Noise Model Instance From QPU	65.915%	96.159%	100%	95.241%	94.484%
FakeBrisbane	64.392%	96.509%	95.241%	100%	93.048%
Noise Model From CSV	67.101%	94.531%	94.484%	93.048%	100%

Table 2: Similarity matrix for ibm_brisbane, five-qubit circuit with depth 10, and optimization level 1

	***ibm_brisbane* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeBrisbane***	**Noise Model From CSV**
***ibm_brisbane* QPU**	100%	89.408%	89.755%	69.577%	91.126%
Simulator Instance From QPU	89.408%	100%	97.849%	77.408%	94.927%
Noise Model Instance From QPU	89.755%	97.849%	100%	77.188%	95.597%
FakeBrisbane	69.577%	77.408%	77.188%	100%	74.326%
Noise Model From CSV	91.126%	94.927%	95.597%	74.326%	100%

Table 3: Similarity matrix for ibm_brisbane, five-qubit circuit with depth 10, and optimization level 2

	***ibm_brisbane* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeBrisbane***	**Noise Model From CSV**
***ibm_brisbane* QPU**	100%	76.453%	76.703%	72.409%	75.109%
Simulator Instance From QPU	76.453%	100%	97.955%	78.057%	94.973%
Noise Model Instance From QPU	76.703%	97.955%	100%	78.593%	94.401%
FakeBrisbane	72.409%	78.057%	78.593%	100%	75.287%
Noise Model From CSV	75.109%	94.973%	94.401%	75.287%	100%

Table 4: Similarity matrix for ibm_brisbane, five-qubit circuit with depth 10, and optimization level 3

	***ibm_brisbane* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeBrisbane***	**Noise Model From CSV**
***ibm_brisbane* QPU**	100%	75.940%	76.119%	70.136%	74.725%
Simulator Instance From QPU	75.940%	100%	98.078%	78.541%	95.114%
Noise Model Instance From QPU	76.119%	98.078%	100%	77.996%	95.532%
FakeBrisbane	70.136%	78.541%	77.996%	100%	75.634%
Noise Model From CSV	74.725%	95.114%	95.532%	75.634%	100%

Table 5: Similarity matrix for ibm_brisbane, five-qubit circuit with depth 20, and optimization level 0

	***ibm_brisbane* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeBrisbane***	**Noise Model From CSV**
***ibm_brisbane* QPU**	100%	92.268%	91.422%	89.546%	92.808%
Simulator Instance From QPU	92.268%	100%	96.976%	95.807%	95.072%
Noise Model Instance From QPU	91.422%	96.976%	100%	95.644%	95.543%
FakeBrisbane	89.546%	95.807%	95.644%	100%	92.609%
Noise Model From CSV	92.808%	95.072%	95.543%	92.609%	100%

Table 6: Similarity matrix for ibm_brisbane, five-qubit circuit with depth 20, and optimization level 1

	***ibm_brisbane* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeBrisbane***	**Noise Model From CSV**
***ibm_brisbane* QPU**	100%	88.925%	88.839%	79.452%	90.643%
Simulator Instance From QPU	88.925%	100%	97.998%	84.713%	95.369%
Noise Model Instance From QPU	88.839%	97.998%	100%	84.798%	95.285%
FakeBrisbane	79.452%	84.713%	84.798%	100%	83.041%
Noise Model From CSV	90.643%	95.369%	95.285%	83.041%	100%

Table 7: Similarity matrix for ibm_brisbane, five-qubit circuit with depth 20, and optimization level 2

	***ibm_brisbane* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeBrisbane***	**Noise Model From CSV**
***ibm_brisbane* QPU**	100%	85.489%	84.772%	71.720%	86.509%
Simulator Instance From QPU	85.489%	100%	97.859%	83.125%	96.255%
Noise Model Instance From QPU	84.772%	97.859%	100%	83.977%	95.752%
FakeBrisbane	71.720%	83.125%	83.977%	100%	82.195%
Noise Model From CSV	86.509%	96.255%	95.752%	82.195%	100%

Table 8: Similarity matrix for ibm_brisbane, five-qubit circuit with depth 20, and optimization level 3

	***ibm_brisbane* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeBrisbane***	**Noise Model From CSV**
***ibm_brisbane* QPU**	100%	85.120%	85.570%	72.263%	87.358%
Simulator Instance From QPU	85.120%	100%	97.123%	83.508%	95.501%
Noise Model Instance From QPU	85.570%	97.123%	100%	84.056%	95.903%
FakeBrisbane	72.263%	83.508%	84.056%	100%	81.618%
Noise Model From CSV	87.358%	95.501%	95.903%	81.618%	100%

Table 9: Similarity matrix for ibm_brisbane, five-qubit circuit with depth 30, and optimization level 0

	***ibm_brisbane* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeBrisbane***	**Noise Model From CSV**
***ibm_brisbane* QPU**	100%	79.123%	80.128%	80.297%	79.426%
Simulator Instance From QPU	79.123%	100%	96.292%	95.272%	97.201%
Noise Model Instance From QPU	80.128%	96.292%	100%	95.003%	95.390%
FakeBrisbane	80.297%	95.272%	95.003%	100%	93.558%
Noise Model From CSV	79.426%	97.201%	95.390%	93.558%	100%

Table 10: Similarity matrix for ibm_brisbane, five-qubit circuit with depth 30, and optimization level 1

	***ibm_brisbane* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeBrisbane***	**Noise Model From CSV**
***ibm_brisbane* QPU**	100%	97.052%	97.203%	89.668%	95.406%
Simulator Instance From QPU	97.052%	100%	97.904%	91.102%	94.850%
Noise Model Instance From QPU	97.203%	97.904%	100%	90.781%	95.394%
FakeBrisbane	89.668%	91.102%	90.781%	100%	88.382%
Noise Model From CSV	95.406%	94.850%	95.394%	88.382%	100%

Table 11: Similarity matrix for ibm_brisbane, five-qubit circuit with depth 30, and optimization level 2

	***ibm_brisbane* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeBrisbane***	**Noise Model From CSV**
***ibm_brisbane* QPU**	100%	95.360%	95.255%	88.554%	93.982%
Simulator Instance From QPU	95.360%	100%	97.782%	91.188%	95.547%
Noise Model Instance From QPU	95.255%	97.782%	100%	91.916%	95.685%
FakeBrisbane	88.554%	91.188%	91.916%	100%	90.275%
Noise Model From CSV	93.982%	95.547%	95.685%	90.275%	100%

Table 12: Similarity matrix for ibm_brisbane, five-qubit circuit with depth 30, and optimization level 3

	***ibm_brisbane* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeBrisbane***	**Noise Model From CSV**
***ibm_brisbane* QPU**	100%	95.901%	96.292%	89.472%	94.654%
Simulator Instance From QPU	95.901%	100%	97.829%	91.116%	96.436%
Noise Model Instance From QPU	96.292%	97.829%	100%	90.943%	95.394%
FakeBrisbane	89.472%	91.116%	90.943%	100%	89.789%
Noise Model From CSV	94.654%	96.436%	95.394%	89.789%	100%

ibm_sherbrooke

Table 13: Similarity matrix for ibm_sherbrooke, five-qubit circuit with depth 10, and optimization level 0

	***ibm_sherbrooke* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeSherbrooke***	**Noise Model From CSV**
***ibm_sherbrooke* QPU**	100%	88.084%	88.215%	87.791%	87.473%
Simulator Instance From QPU	88.084%	100%	97.603%	96.111%	94.037%
Noise Model Instance From QPU	88.215%	97.603%	100%	97.256%	93.410%
FakeSherbrooke	87.791%	96.111%	97.256%	100%	91.466%
Noise Model From CSV	87.473%	94.037%	93.410%	91.466%	100%

Table 14: Similarity matrix for ibm_sherbrooke, five-qubit circuit with depth 10, and optimization level 1

	***ibm_sherbrooke* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeSherbrooke***	**Noise Model From CSV**
***ibm_sherbrooke* QPU**	100%	64.504%	64.510%	68.144%	73.614%
Simulator Instance From QPU	64.504%	100%	98.004%	76.737%	86.542%
Noise Model Instance From QPU	64.510%	98.004%	100%	77.041%	86.612%
FakeSherbrooke	68.144%	76.737%	77.041%	100%	73.459%
Noise Model From CSV	73.614%	86.542%	86.612%	73.459%	100%

Table 15: Similarity matrix for ibm_sherbrooke, five-qubit circuit with depth 10, and optimization level 2

	***ibm_sherbrooke* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeSherbrooke***	**Noise Model From CSV**
***ibm_sherbrooke* QPU**	100%	70.587%	70.324%	75.682%	75.683%
Simulator Instance From QPU	70.587%	100%	98.196%	77.886%	89.654%
Noise Model Instance From QPU	70.324%	98.196%	100%	77.918%	89.363%
FakeSherbrooke	75.682%	77.886%	77.918%	100%	72.537%
Noise Model From CSV	75.683%	89.654%	89.363%	72.537%	100%

Table 16: Similarity matrix for ibm_sherbrooke, five-qubit circuit with depth 10, and optimization level 3

	***ibm_sherbrooke* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeSherbrooke***	**Noise Model From CSV**
***ibm_sherbrooke* QPU**	100%	71.035%	71.564%	76.762%	76.053%
Simulator Instance From QPU	71.035%	100%	98.062%	78.321%	88.973%
Noise Model Instance From QPU	71.564%	98.062%	100%	78.241%	89.544%
FakeSherbrooke	76.762%	78.321%	78.241%	100%	73.784%
Noise Model From CSV	76.053%	88.973%	89.544%	73.784%	100%

Table 17: Similarity matrix for ibm_sherbrooke, five-qubit circuit with depth 20, and optimization level 0

	***ibm_sherbrooke* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeSherbrooke***	**Noise Model From CSV**
***ibm_sherbrooke* QPU**	100%	85.749%	86.318%	89.600%	83.867%
Simulator Instance From QPU	85.749%	100%	97.112%	94.827%	93.552%
Noise Model Instance From QPU	86.318%	97.112%	100%	95.283%	93.805%
FakeSherbrooke	89.600%	94.827%	95.283%	100%	91.993%
Noise Model From CSV	83.867%	93.552%	93.805%	91.993%	100%

Table 18: Similarity matrix for ibm_sherbrooke, five-qubit circuit with depth 20, and optimization level 1

	***ibm_sherbrooke* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeSherbrooke***	**Noise Model From CSV**
***ibm_sherbrooke* QPU**	100%	83.506%	83.626%	72.516%	88.074%
Simulator Instance From QPU	83.506%	100%	98.110%	80.585%	91.060%
Noise Model Instance From QPU	83.626%	98.110%	100%	80.712%	90.556%
FakeSherbrooke	72.516%	80.585%	80.712%	100%	74.279%
Noise Model From CSV	88.074%	91.060%	90.556%	74.279%	100%

Table 19: Similarity matrix for ibm_sherbrooke, five-qubit circuit with depth 20, and optimization level 2

	***ibm_sherbrooke* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeSherbrooke***	**Noise Model From CSV**
***ibm_sherbrooke* QPU**	100%	81.743%	81.838%	73.167%	86.469%
Simulator Instance From QPU	81.743%	100%	98.349%	80.804%	90.011%
Noise Model Instance From QPU	81.838%	98.349%	100%	80.810%	90.630%
FakeSherbrooke	73.167%	80.804%	80.810%	100%	73.973%
Noise Model From CSV	86.469%	90.011%	90.630%	73.973%	100%

Table 20: Similarity matrix for ibm_sherbrooke, five-qubit circuit with depth 20, and optimization level 3

	***ibm_sherbrooke* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeSherbrooke***	**Noise Model From CSV**
***ibm_sherbrooke* QPU**	100%	78.884%	78.332%	81.031%	81.036%
Simulator Instance From QPU	78.884%	100%	97.990%	76.219%	92.806%
Noise Model Instance From QPU	78.332%	97.990%	100%	75.887%	92.602%
FakeSherbrooke	81.031%	76.219%	75.887%	100%	72.680%
Noise Model From CSV	81.036%	92.806%	92.602%	72.680%	100%

Table 21: Similarity matrix for ibm_sherbrooke, five-qubit circuit with depth 30, and optimization level 0

	***ibm_sherbrooke* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeSherbrooke***	**Noise Model From CSV**
***ibm_sherbrooke* QPU**	100%	90.561%	90.525%	91.449%	88.106%
Simulator Instance From QPU	90.561%	100%	97.978%	97.293%	92.400%
Noise Model Instance From QPU	90.525%	97.978%	100%	96.564%	93.388%
FakeSherbrooke	91.449%	97.293%	96.564%	100%	91.761%
Noise Model From CSV	88.106%	92.400%	93.388%	91.761%	100%

Table 22: Similarity matrix for ibm_sherbrooke, five-qubit circuit with depth 30, and optimization level 1

	***ibm_sherbrooke* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeSherbrooke***	**Noise Model From CSV**
***ibm_sherbrooke* QPU**	100%	85.439%	85.482%	75.472%	90.487%
Simulator Instance From QPU	85.439%	100%	98.210%	78.858%	91.951%
Noise Model Instance From QPU	85.482%	98.210%	100%	78.678%	92.102%
FakeSherbrooke	75.472%	78.858%	78.678%	100%	76.867%
Noise Model From CSV	90.487%	91.951%	92.102%	76.867%	100%

Table 23: Similarity matrix for ibm_sherbrooke, five-qubit circuit with depth 30, and optimization level 2

	***ibm_sherbrooke* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeSherbrooke***	**Noise Model From CSV**
***ibm_sherbrooke* QPU**	100%	92.917%	92.502%	80.484%	93.549%
Simulator Instance From QPU	92.917%	100%	98.129%	76.485%	94.022%
Noise Model Instance From QPU	92.502%	98.129%	100%	76.287%	93.595%
FakeSherbrooke	80.484%	76.485%	76.287%	100%	75.289%
Noise Model From CSV	93.549%	94.022%	93.595%	75.289%	100%

Table 24: Similarity matrix for ibm_sherbrooke, five-qubit circuit with depth 30, and optimization level 3

	***ibm_sherbrooke* QPU**	**Simulator Instance From QPU**	**Noise Model Instance From QPU**	***FakeSherbrooke***	**Noise Model From CSV**
***ibm_sherbrooke* QPU**	100%	92.810%	91.594%	80.149%	92.389%
Simulator Instance From QPU	92.810%	100%	97.583%	76.008%	94.054%
Noise Model Instance From QPU	91.594%	97.583%	100%	74.463%	94.299%
FakeSherbrooke	80.149%	76.008%	74.463%	100%	74.009%
Noise Model From CSV	92.389%	94.054%	94.299%	74.009%	100%